Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<th inking process >
Step 1 : Understand the initial cost .
The tree costs $ 9 0 to plant . This is a one -time cost .

Step 2 : Determine the annual income and expenses .
Each year , the tree produces 7 le mons .
Each lemon sells for $ 1 . 5 , so annual revenue = 7 ÃĹ $ 1 . 5 = $ 1 0 . 5 0 .
Annual cost to water and feed = $ 3 .

So , annual net profit = Revenue - Cost = $ 1 0 . 5 0 - $ 3 = $ 7 . 5 0 per year .

Step 3 : We want to find out how many years it takes before Carlos starts earning money .
Since the initial cost is $ 9 0 and each year he earns a net profit of $ 7 . 5 0 , we need to find when the cumulative profit covers the initial cost .

So , we need to solve :
Total net profit over Y years âī¥ $ 9 0

Total net profit = Y ÃĹ $ 7 . 5 0

We want :
Y ÃĹ 7 . 5 0 âī¥ 9 0

So :
Y âī¥ 9 0 / 7 . 5 0 = 1 2

Therefore , after 1 2 years , Carlos will have earned enough to cover the initial cost , and from year 1 3 onwards , he will be earning a profit .

But the question is : " How many years will it take before he starts earning money ?"

" Start s earning money " means the point when the cumulative profit covers the initial cost âĢĶ i .e ., when he is no longer losing money .

So , at the end of year 1 2 , the total profit is 1 2 ÃĹ 7 . 5 0 = $ 9 0 , which just covers the initial $ 9 0 .

So , after 1 2 years , he breaks even âĢĶ and from year 1 3 onward , he starts earning money .

But " before he starts earning money " âĢĶ does this mean when he starts making a profit (i .e ., positive net profit), or when the cost is covered ?

Typ ically , " starts earning money " means when he begins to make a profit âĢĶ i .e ., when the cumulative profit covers the initial cost .

So , after 1 2 years , he has broken even . He starts earning money from year 1 3 .

But the question is : " how many years will it take before he starts earning money ?"

This could be interpreted as : after how many years does he begin to make a profit ?

So , after 1 2 years , he has just covered the cost âĢĶ he starts earning money from year 1 3 .

But if we interpret " starts earning money " as the year in which the net profit becomes positive , then after 1 2 years , the total profit is $ 9 0 , which just covers the cost âĢĶ so the net profit is zero .

So , he has not yet started earning money âĢĶ he is breaking even .

To start earning money , he must have a positive net profit .

So , when does he have a positive net profit ?

After year 1 2 , he has broken even âĢĶ so no net profit .

After year 1 3 , he has a net profit .

But the question is : " how many years will it take before he starts earning money ?"

So , after how many years ?

So , after 1 2 years , he has just covered the cost âĢĶ so he starts earning money from year 1 3 .

But the number of years it takes to start earning money âĢĶ is it 1 2 or 1 3 ?

Let âĢĻs clarify with a real -world interpretation .

" Start s earning money " âĢĶ when does he begin to make a profit ?

After 1 2 years , he has spent $ 9 0 and earned $ 9 0 âĢĶ so he is breaking even .

He does not earn money until after that .

So , after 1 3 years , he has earned $ 1 0 . 5 0 ÃĹ 1 3 = $ 1 3 6 . 5 0 , minus $ 3 ÃĹ 1 3 = $ 3 9 , so net profit = $ 9 7 . 5 0